

What is NLP?

Natural Language Processing, or *NLP*, is the sub-field of AI that is focused on enabling computers to understand and process human languages.

Need for NLP (Why NLP?)

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables machines to understand human language. Its goal is to build systems that can make sense of the text and automatically perform tasks like translation, spell check, or topic classification. NLP has changed the way we interact with computers and it'll continue doing so in the future. These AI technologies will be the underlying force for transformation from data-driven to intelligence-driven endeavors, as they shape and improve communication technology in the years to come.

Applications of NLP

Real-World Applications of NLP

- Contextual Advertisements
- Email Clients — spam filtering, smart reply
- Social Media — removing adult content, hate speech, opinion mining
- Search Engines
- Chatbots

Common NLP tasks

Text/Document Classification-By classifying text, we are aiming to assign one or more classes or categories to a document, making it easier to manage and sort. This is especially useful for publishers, news sites, blogs, or anyone who deals with a lot of content.

Sentiment Analysis(opinion mining)-identifies emotion tone behind a body of text. This is a popular way for organizations to determine and categorize opinions about a product, service, or idea.

Information Retrieval-defined as a software program that deals with the organization, storage, retrieval, and evaluation of information from document repositories, particularly textual information. The system assists users in finding the information they require but it does not explicitly return the answers to the questions. It informs the existence and location of documents that might consist of the required information

Part of Speech Tagging- is a popular Natural Language Processing process that refers to categorizing words in a text (corpus) in correspondence with a particular part of speech, depending on the definition of the word and its context

Language Detection and Machine Translation- Machine translation is the process of translating a text in one natural language into another natural language using a computer system. Translating a document containing a single source language contents is easy but when the information in the source document is given in multilingual format then there is a need to identify the languages that are involved in such multilingual document. Language identification is the task in natural language processing that automatically identifies the natural language in which the content in the given document is written. Language identification is the fundamental and crucial step in many NLP applications.

Conversational Agents- also known as chatbots, are computer programs designed to simulate human text or verbal conversations

Knowledge Graph and QA Systems- An automated question answering system applies a set of NLP techniques to analyze unstructured documents — from Wikipedia articles to social media newsfeed or medical records — by retrieving the needed information piece, analyzing it, and using the best part to answer the question

Text Summarization- Text summarization is the technique for generating a concise and precise summary of voluminous texts while focusing on the sections that convey useful information, and without losing the overall meaning

Topic Modelling- Topic Modelling is the task of using unsupervised learning to extract the main topics (represented as a set of words) that occur in a collection of documents.

Text Generation- Text generation is a subfield of natural language processing. It leverages knowledge in computational linguistics and artificial intelligence to automatically generate natural language texts, which can satisfy certain communicative requirements.

Spell Checking and Grammar Correction- The spell checker compares every word typed with a list of thousands of correctly spelled words and then uses algorithms to determine the correct spellings

Text Parsing- Text Parsing is the process of analyzing text made of a sequence of tokens to determine its grammatical structure concerning a given (more or less) formal grammar.

Speech to Text- Natural Language Processing speech to text is a profound application of Deep Learning which allows the machines to understand human language and read it with a motive to act and react, as usual, humans do.